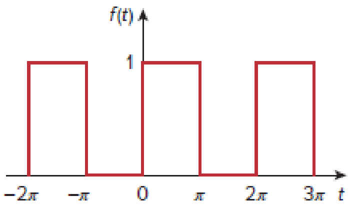
Problem

The complex Fourier series of the function in Fig. 17.83(a) is

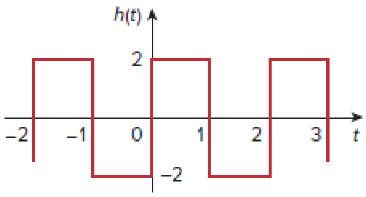
$$f(t)=\frac{1}{2}-\sum_{n=-\infty}^{\infty}\frac{je^{-j(2n+1)t}}{(2n+1)\pi}$$

Find the complex Fourier series of the function $h(t)$in Fig. 17.83(b).

Figure 17.83:



(a)

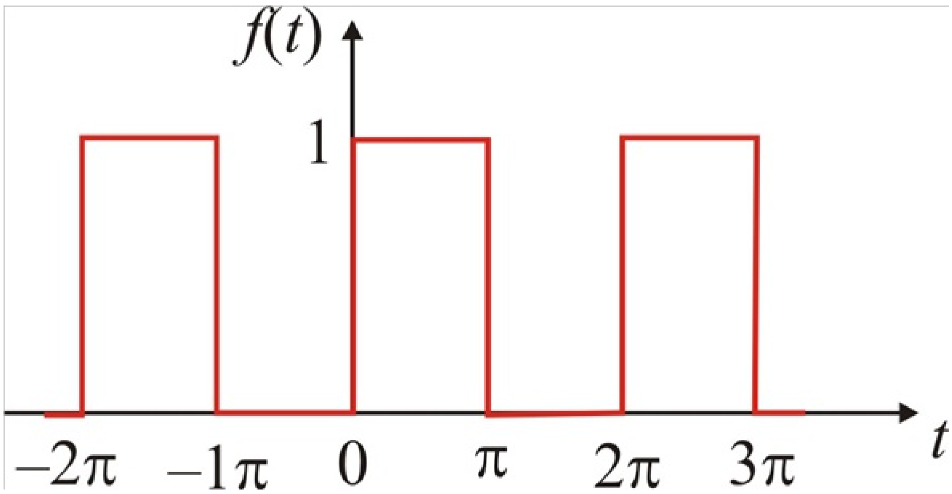


(b)

Step-by-step solution

Step 1 of 8

Given function



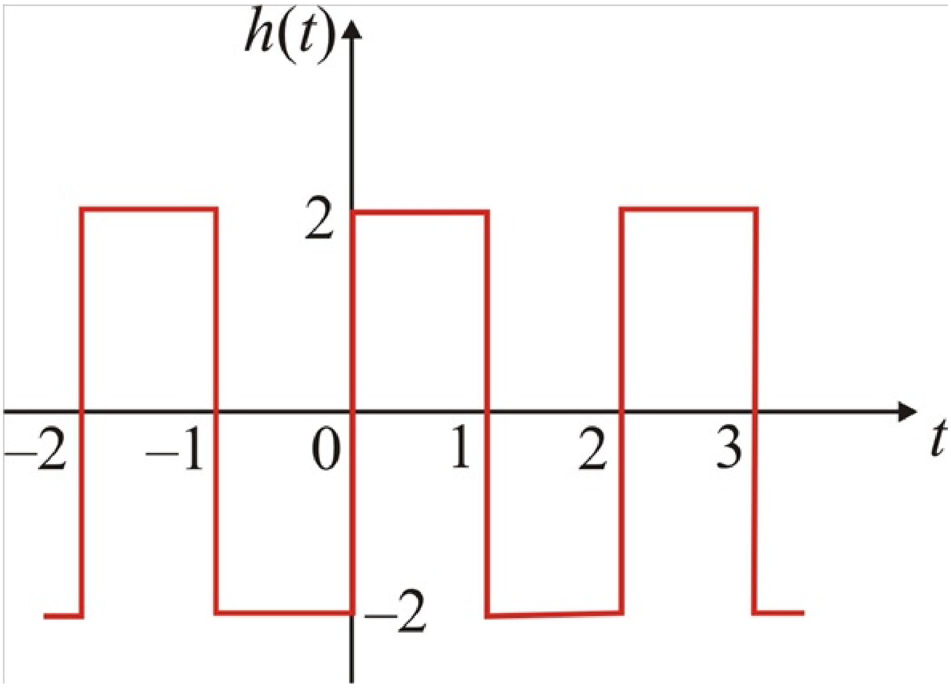
Step 2 of 8

Fourier series of above function

$$f(t)=\frac{1}{2}-\sum_{n=-\infty}^{\infty}\frac{je^{-(2n+1)t}}{(2n+1)\pi}$$

Step 3 of 8

Function $h(t)$ is



Step 4 of 8

Function $f(t)$ is defined as

$$f(t)=\begin{cases}1 & \text{for } 0<t<\pi \\ 0 & \text{for } \pi<t<2\pi\end{cases}$$

And function $h(t)$ is defined as

$$h(t)=\begin{cases}2 & \text{for } 0<t<1 \\ -2 & \text{for } 1<t<2\end{cases}$$

Step 5 of 8

Time period of function $f(t)$ is

$$T=2\pi$$

We know that $\omega_o=\frac{2\pi}{T}$

$$\omega_o=\frac{2\pi}{2\pi}$$

$$\omega_o=1$$

Time period of function $h(t)$ is

$$T=2$$

We know that $\omega_o=\frac{2\pi}{T}$

$$\omega_o=\frac{2\pi}{2}$$

$$\omega_o=\pi$$

Step 6 of 8

DC component for function $f(t)$ is

$$a_0 = \frac{(1\times\pi)+(0\times\pi)}{2\pi}$$

$$a_0 = \frac{1}{2}$$

DC component for function $h(t)$ is

$$a_0 = \frac{(2\times1)+(-2\times1)}{2\pi}$$

$$a_0 = 0$$

Therefore dc component for $h(t)$ is zero.

Step 7 of 8

Complex Fourier coefficient for $h(t)$

$$C_n = \frac{1}{2}\int_0^2 h(t)e^{-jn\pi t}dt$$

$$C_n = \frac{1}{2}\left[\int_0^1 (2)e^{-jn\pi t}dt + \int_1^2 (-2)e^{-jn\pi t}dt\right]$$

$$C_n = \frac{1}{2}\left[2\left[\frac{e^{-jn\pi t}}{-jn\pi}\right]_0^1 - 2\left[\frac{e^{-jn\pi t}}{-jn\pi}\right]_1^2\right]$$

$$C_n = \frac{1}{-jn\pi}\left[\left[e^{-jn\pi t}\right]_0^1 - \left[e^{-jn\pi t}\right]_1^2\right]$$

$$C_n = \frac{1}{-jn\pi}\left[\left(e^{-jn\pi}-1\right)-\left(e^{-j2n\pi}-e^{-jn\pi}\right)\right]$$

$$C_n = \frac{1}{-jn\pi}\left(-e^{-j2n\pi}+2e^{-jn\pi}-1\right)$$

$$C_n = \frac{1}{jn\pi}\left(e^{-j2n\pi}-2e^{-jn\pi}+1\right)$$

$$C_n = \frac{1}{jn\pi}\left(1-2(-1)^n+1\right) \qquad \left(\begin{array}{l}\because e^{-jn\pi} = \cos n\pi - j\sin n\pi = (-1)^n \\ e^{-j2n\pi} = \cos 2n\pi - j\sin 2n\pi = 1\end{array}\right)$$

$$C_n = \frac{2}{jn\pi}\left(1-(-1)^n\right)$$

$$C_n = \begin{cases}\dfrac{4}{jn\pi} & \text{for } n = \text{odd} \\ 0 & \text{for } n = \text{even}\end{cases}$$

Step 8 of 8

Therefore required function $h(t)$ is

$$\boxed{f(t) = -\sum_{\substack{n=-\infty \\ n\neq0}}^{\infty} \frac{j4e^{j(2n+1)\pi t}}{(2n+1)\pi}}$$